LAB5 Feature Tracking Comparison

VNAV2021 code | Ubuntu 20.04 / ROS Noetic | OpenCV 3.4.2

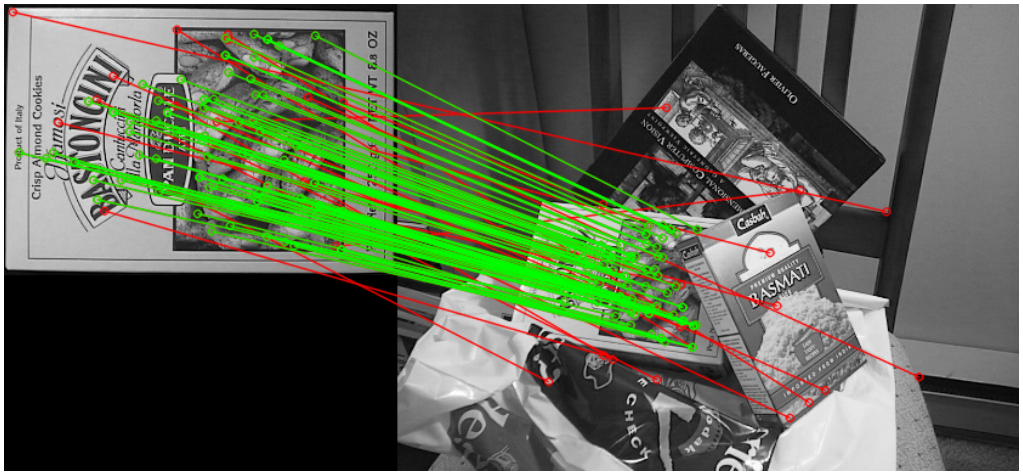
Methods and scoring

SIFT and SURF use FLANN two-neighbor matching and the 0.8 distance ratio. ORB uses Hamming-distance brute force with the same ratio. Harris+LK tracks corners by pyramidal optical flow. All methods use fundamental-matrix RANSAC with a 3-pixel threshold and 0.99 confidence to count geometric inliers.

Bundled image pair

box.png to box_in_scene.png. SIFT descriptors have 128 float values; SURF has 64 float values; ORB has 32 bytes.

Method	Keypoints 1	Keypoints 2	Raw matches	Good matches	Inliers	Inlier ratio
SIFT	603	969	603	99	79	79.8%
SURF	997	1822	997	129	85	65.9%
ORB	500	500	500	13	7	53.8%

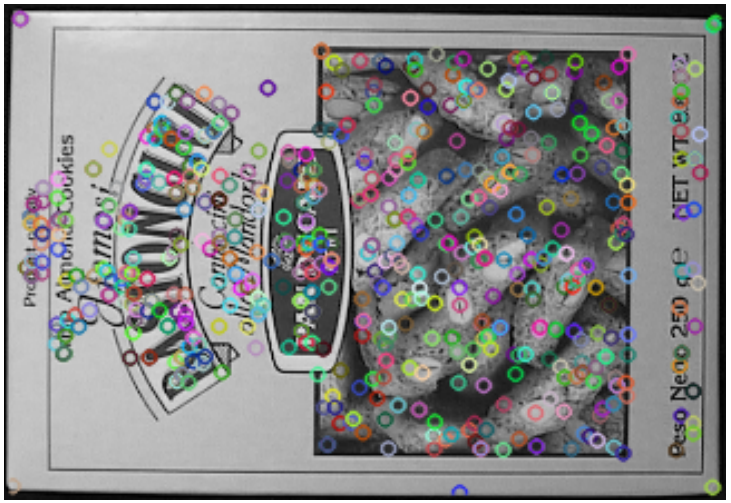


Observation. The large viewpoint and scale change makes this pair harder than adjacent video frames. SURF gave the most RANSAC inliers (85), while SIFT had the higher inlier fraction (79.8%). At the supplied 500-feature, one-level setting, ORB retained only 13 unambiguous matches and 7 inliers. Green lines in the figure are inliers; red lines are rejected matches.

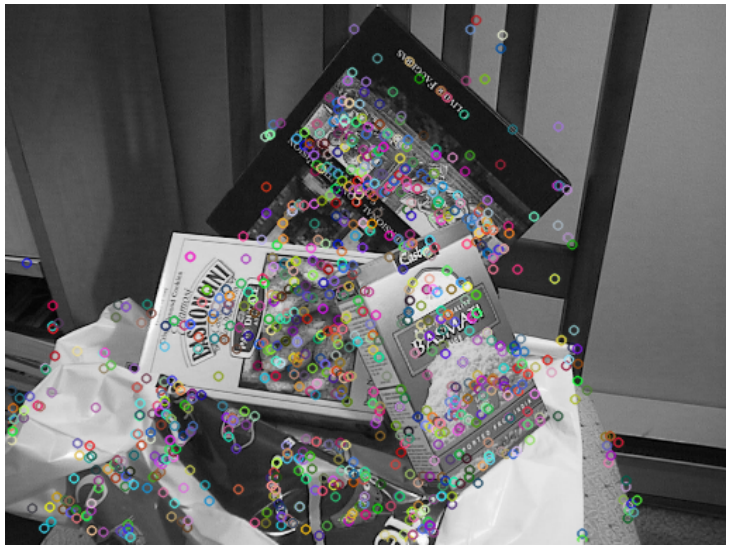
SIFT detection and matching figures

The two-image run saves these figures directly from the tracker. There are 603 and 969 keypoints, 603 raw nearest-neighbor matches, and 99 matches after the ratio test.

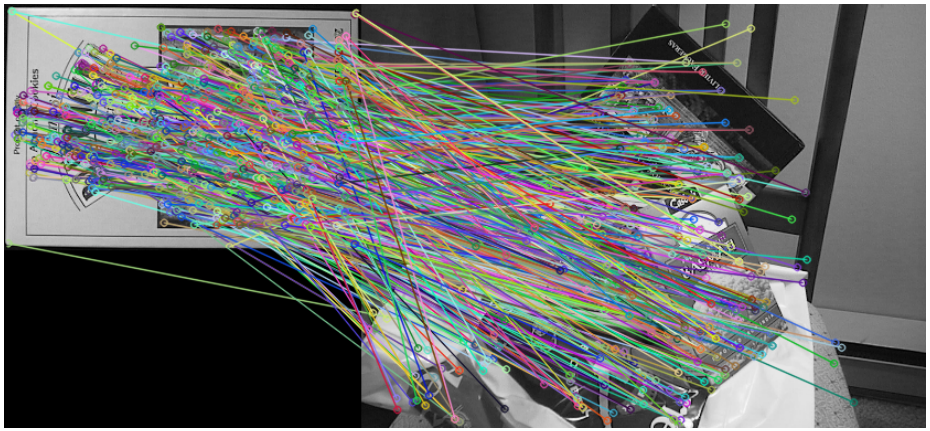
1. SIFT keypoints in box.png



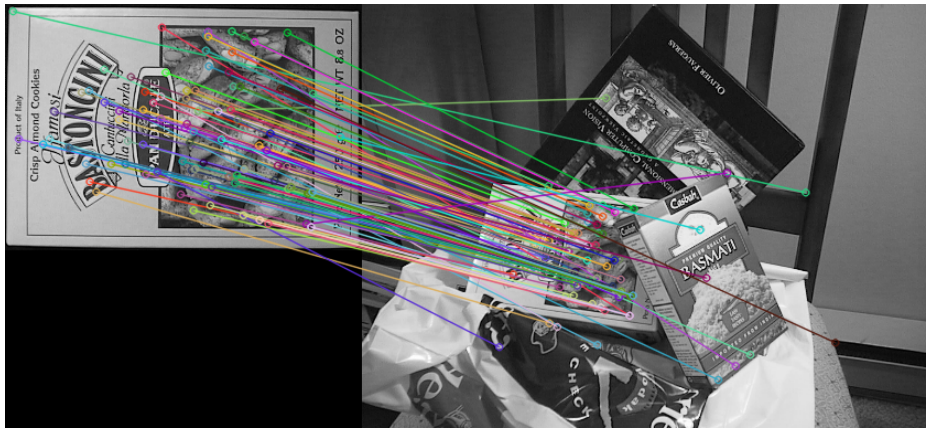
2. SIFT keypoints in box_in_scene.png



3. Raw nearest-neighbor matches



4. Matches after the 0.8 ratio test



The ratio test removes ambiguous correspondences before geometric verification. The colored inlier/outlier plot appears on page 1.

Real flight sequences

Each table averages all adjacent RGB-frame pairs. Avg. ratio is the mean per-pair inlier fraction. Playback was slowed as needed so the VM did not drop images.

Dataset 1: 30fps room sequence

30fps_424x240_2018-10-01-18-35-06.bag | 1584 RGB frames | 0.6x playback

Method	Avg. KP1	Avg. KP2	Avg. matches	Avg. good	Avg. inliers	Avg. ratio	Pairs
SIFT	403.5	403.6	403.5	258.4	249.0	95.7%	1583
SURF	466.9	467.0	466.9	334.2	319.9	95.3%	1583
ORB	315.7	315.7	315.7	230.2	226.0	97.9%	1583
Harris+LK	102.2	102.2	100.9	100.9	100.8	99.9%	1583

Dataset 2: smooth trajectory

vnav-lab5-smooth-trajectory.bag | 1386 RGB frames | 0.35x SIFT/SURF; 0.6x ORB/LK

Method	Avg. KP1	Avg. KP2	Avg. matches	Avg. good	Avg. inliers	Avg. ratio	Pairs
SIFT	147.8	147.9	147.8	92.9	80.7	86.1%	1385
SURF	1191.8	1192.9	1191.8	714.1	609.6	85.5%	1385
ORB	278.8	279.1	278.8	166.8	151.2	90.8%	1385
Harris+LK	70.6	70.7	68.9	68.9	64.2	93.9%	1385

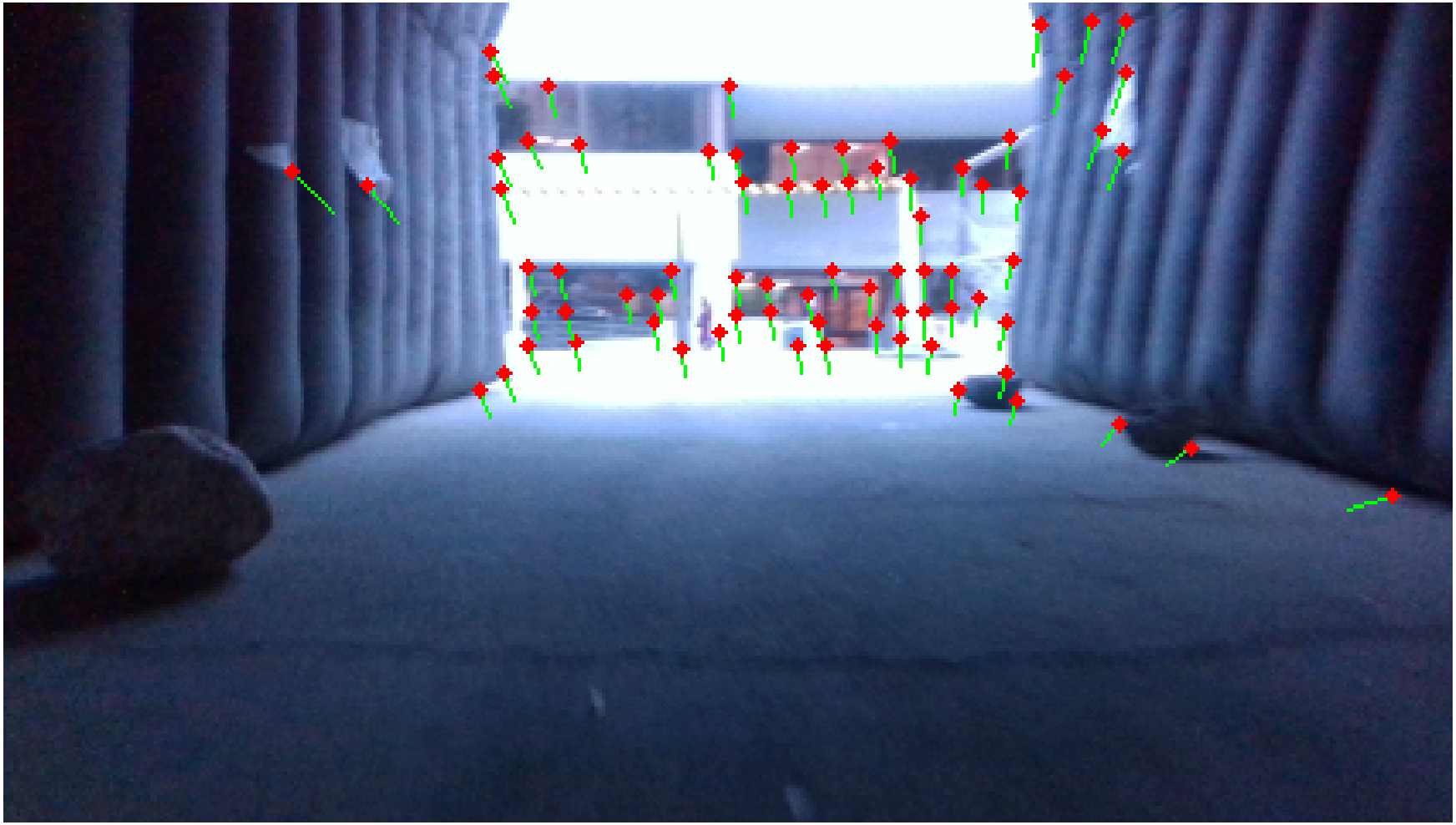
Interpretation and limits

SURF produced the most average inliers on both real sequences (320 and 610). ORB was weak on the separated box pair, yet exceeded SIFT on the smooth trajectory (151 versus 81 inliers), showing the effect of scene and motion. Harris+LK had the highest inlier fraction but fewer tracks. It assumes small local displacement and roughly constant brightness; occlusion and appearance changes can break tracks. Valid tracks count as good matches, with no descriptor ratio test. Runtime was not measured, so no speed ranking is claimed.

For the LK visual, see the next page. The ROS Noetic cv_bridge links system OpenCV 4.2 while the historical lab builds OpenCV 3.4.2; both modes ran on this VM despite linker warnings.

Harris + LK track example

Real RGB frames 300 and 305 from Dataset 1. The five-frame gap makes short tracks visible; 75 of 78 previous-frame corners were tracked in this illustration.



Green segments show motion between the two frames; red dots mark tracked positions in the later image. The full-dataset LK statistics on page 3 use consecutive frames, as do the other methods.

Assignment and data: MIT 16.485 VNAV LAB5 Exercises (2023), vnav.mit.edu/labs_2023/lab5/exercises.html; github.com/MIT-SPARK/VNAV-lab-data.